

MAX MATTHEW

Vehicle Dynamics Engineer

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PROFESSIONAL SUMMARY

First-class Motorsport Engineering graduate and MRes candidate in Vehicle Dynamics, combining OEM vehicle evaluation experience at Nissan Technical Centre Europe with Formula Student leadership as Team Principal. Skilled in translating multibody simulation (Adams Car, MATLAB) into validated, trackside performance decisions from chassis geometry development through physical rig correlation.

EDUCATION

Master's by Research (MRes) – Vehicle Dynamics | University of Huddersfield | Oct 2026 – Oct 2027

- Thesis:** *Numerical and Experimental Evaluation of Suspension System for Formula Student Racing Car* (Supported by Vice-Chancellor's full fee-waiver scholarship).
- Developing a predictive optimisation tool for pre-event dynamics tuning, validated by physical K&C testing via 4-post rig and track data.
- Modelling a roll-decoupled suspension in Adams Car and linking variable parameters (spring stiffness, roll stiffness etc.) via a MATLAB lap simulation script.

BEng (Hons) - Automotive & Motorsport Engineering | University of Huddersfield | Sep 2022 – Jul 2026

- Result: First-Class Honours (80.7%).** Key Modules: Vehicle Handling & Performance Prediction (93%), Aerodynamics & CFD (80%).
- FYP:** *Design & analysis of a roll-decoupled suspension system for a Formula Student vehicle* (85%, IMechE Best Project Award).

RELEVANT EXPERIENCE

Nissan Technical Centre Europe | Vehicle Evaluation & Test Management | Aug 2024 – Aug 2025

- Evaluated production passenger vehicles, analysing vehicle dynamics, NVH behaviour and subjective customer feedback to identify chassis-related development concerns.
- Developed MATLAB tools to automate vehicle-test data processing and reporting, reducing analysis time by >90% across 100+ development vehicles and accelerating engineering concern review
- Built Excel dashboards tracking vehicle concerns through the V-model, enabling earlier failure-mode identification and prioritisation of repeat-test requirements across 100+ vehicles
- Presented quantitative vehicle-performance findings to European and Japanese R&D leadership, translating test data into engineering recommendations during joint development reviews.

Team HARE Formula Student | University of Huddersfield | Feb 2023 – Present

- CV Team Principal & Head of Vehicle Dynamics** | Aug 2025 – Jul 2026
- Directed a core team of 6 engineers to successfully pass full FSUK scrutineering and compete in dynamic track events, overcoming severe resource and timeline constraints.
- Led a full suspension redesign, developing custom front U- and rear T-section anti-roll bars; correlated suspension kinematics in Adams Car against design targets for roll centre, anti-geometry and bump steer.
- Lap Simulation Sub-Team Lead** | Feb 2023 – Jul 2024
- Produced DFMA-compliant 2D and 3D SolidWorks assembly drawings for the HARE24 4WD EV (suspension, bodywork, and aero packages) including tolerance analysis and GD&T compliance.
- Built and ran a full-vehicle lap simulation model in IPG CarMaker for the Formula Student Lap Time Simulation competition, optimising predicted lap time through iterative tuning of vehicle parameters (mass distribution, aero, powertrain, suspension setup).

HGP Racing Greenpower F24+ | Team Principal & Performance Engineer | Feb 2023 – Aug 2024

- Built MATLAB strategy simulation tools and optimized a low-drag aero package, improving lap time prediction accuracy by 40% and achieving 3rd place at the International Final.

PERSONAL PROJECTS

MATLAB Suspension Load Solver & Structural Analysis Tool

- Developed a MATLAB suspension load solver to automate 360° GG-sweep load cases, calculating worst-case tension, compression and bending loads for structural sizing of multi-link and trailing-arm systems.
- Evaluated structural failure modes by calculating critical safety factors and adhesive bond overlaps for carbon-to-aluminium nodes, validating theoretical shear limits against physical Instron pull testing.

Double-Wishbone Geometry Calculator & Visualiser

- Developed an interactive suspension kinematics tool in Excel (roll centre height, KPI, caster, scrub radius, motion ratios) from user-defined hardpoints, with real-time 2D visualisation of camber, roll centre migration, and dive/squat behaviour under load.
- Calculated load transfer, wheel rates, ride frequencies, roll stiffnesses as basis for suspension design. Model correlated in Adams Car

TECHNICAL SKILLS

- Vehicle Dynamics & Simulation:** Adams Car, IPG CarMaker, SolidWorks Motion, lap-time simulation, suspension kinematics, vehicle handling, Kinematics & Compliance.
- Programming & Data:** MATLAB/Simulink, Python, Excel/Power Query, CANalyzer, CANape, data analysis, signal processing.
- Testing & Instrumentation:** CAN/LIN, Vector VN1670/VP6450, MoTeC M800, vehicle data acquisition, strain gauges, thermocouples, K&C testing, 4-post rig testing
- CAD / FEA:** SolidWorks, Siemens NX, AutoCAD, Ansys, SolidWorks Simulation, GD&T, DFMA, FMEA

ACHIEVEMENTS, CERTIFICATIONS & INTERESTS

- Certifications:** Certified SOLIDWORKS Associate (CSWA).
- Courses:** Complete Guide to Vehicle Dynamics in F1 – Racetrack Dynamics
- Awards:** IMechE Best Project Award; 2nd Place – IMechE ADYC Automotive Engineering Competition; 3rd Place – Greenpower F24+ International Final.
- Leadership:** President, Hudds Motorsport; Founder/President, NTCE motorsport club, BUKC Clubman competitor.

References available upon request